Spaced retrieval

Revisiting knowledge over time

Use spaced retrieval when students can answer soon after teaching but struggle weeks later. Plan repeated returns to important knowledge rather than treating a topic as finished.

How to do it well

1 Identify the durable knowledge. Select the curriculum content students will need to use again, especially prerequisites for later topics.

2 Map return points. Include an initial practice opportunity and later retrieval across lessons, quizzes and homework.

3 Vary the cues carefully. Ask for the same knowledge through a definition, example, diagram or application without suddenly testing untaught content.

4 Respond to forgetting. Explain the important error, provide an accurate model and give another independent retrieval opportunity.

5 Adjust the next return. Revisit fragile items sooner and maintain occasional retrieval of secure knowledge; do not use a rigid interval regardless of success.

What this sounds like

“This question returns to a topic from three weeks ago. Try it from memory before using the worked solution.”

Classroom example. An English team maps several returns to important terminology across a unit. Pupils first retrieve definitions, later identify examples and later explain the term in a new text. Incorrect responses receive feedback before another attempt. The returns are spaced across learning rather than concentrated into a single revision lesson before the assessment.

Top tips that make the technique work

Space the learning, not just the timetable. A later prompt should require students to recall the idea again; displaying the answer beforehand removes that opportunity.

Expect effort. Some forgetting makes retrieval demanding, but repeated near-total failure suggests the delay or prompt needs adjusting.

Keep a curriculum record. Track what has been retrieved and which ideas need another return so selection is deliberate.

Combine spacing with feedback. Correct errors before the next attempt; do not repeatedly rehearse the same wrong answer.

Distinguish spacing from interleaving. Spacing concerns time between encounters; interleaving concerns mixing problem types that require selection.

Do and avoid

Do	Avoid
Plan returns across lessons and units.	Leaving all revision until the end of a unit.
Include both recent and older important knowledge.	Assuming one delayed quiz is sufficient for every student.

Further reading

Practitioner guidance; teacher language above is an illustrative BSCS adaptation.

DistillED - Successive Relearning | Barton - Four key retrieval opportunities